

FINAL PROJECT

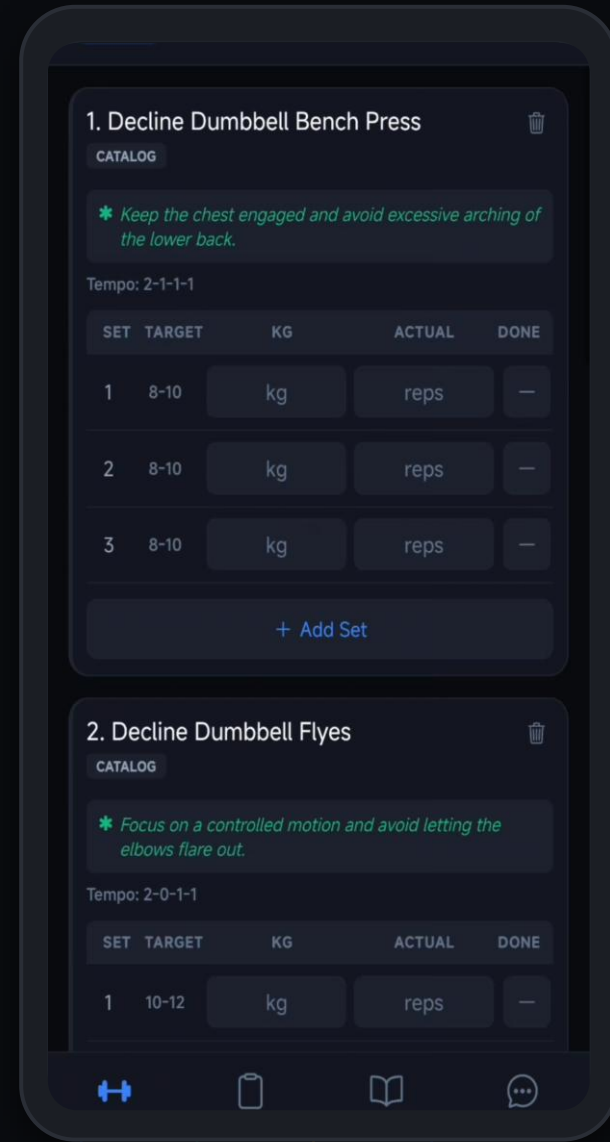
AI GYM COACH

An AI-Powered Personal Training & Fitness Intelligence App

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THE PROBLEM

Fitness Advice Is Everywhere. Personalization Isn't.

Generic Influencer Tips

One-Size Templates

Conflicting Forum Advice

Copy-Paste PDFs

Random Split Routines

Unverified Chatbot Answers

NOISE, NOT GUIDANCE



TRUE PERSONALIZATION

Your body. Your equipment. Your pain. Your recovery.
Still missing from mainstream fitness apps.

MOTIVATION

Why This Idea?



Personalization



Safety



Accessibility



Data-Driven Coaching

THE EVOLUTION

Static Apps

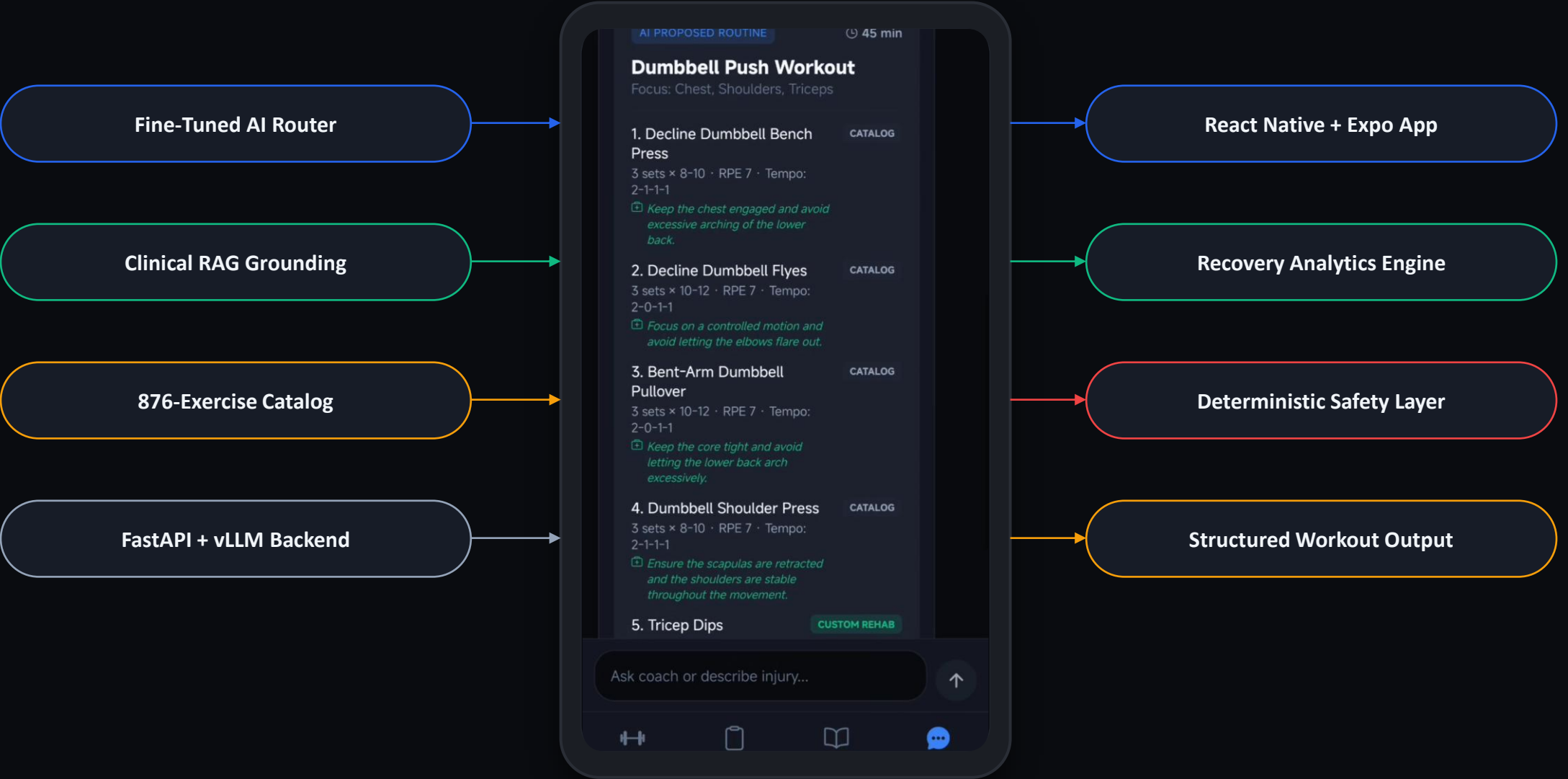


Chatbots

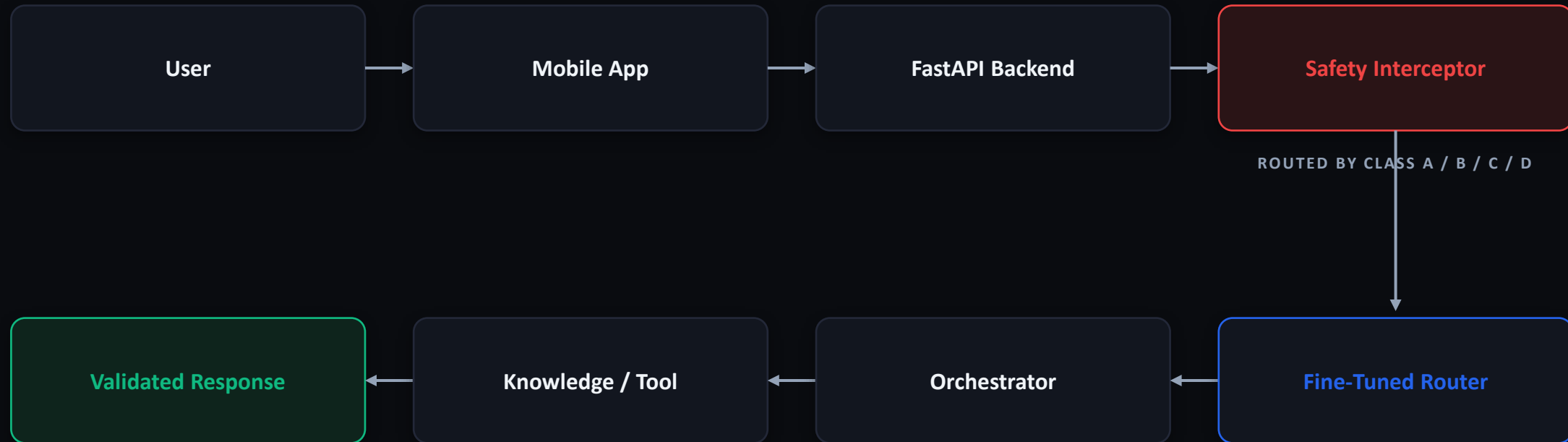


AI Gym Coach

What We Built



The Complete System



Grounded intelligence, controlled by deterministic software at every step.

The Data Behind the System

967

Curated routing records

876

Structured exercises

40

Clinical knowledge cards

768

Embedding dimensions

5

Periodization prompt assets

Teaching the Router

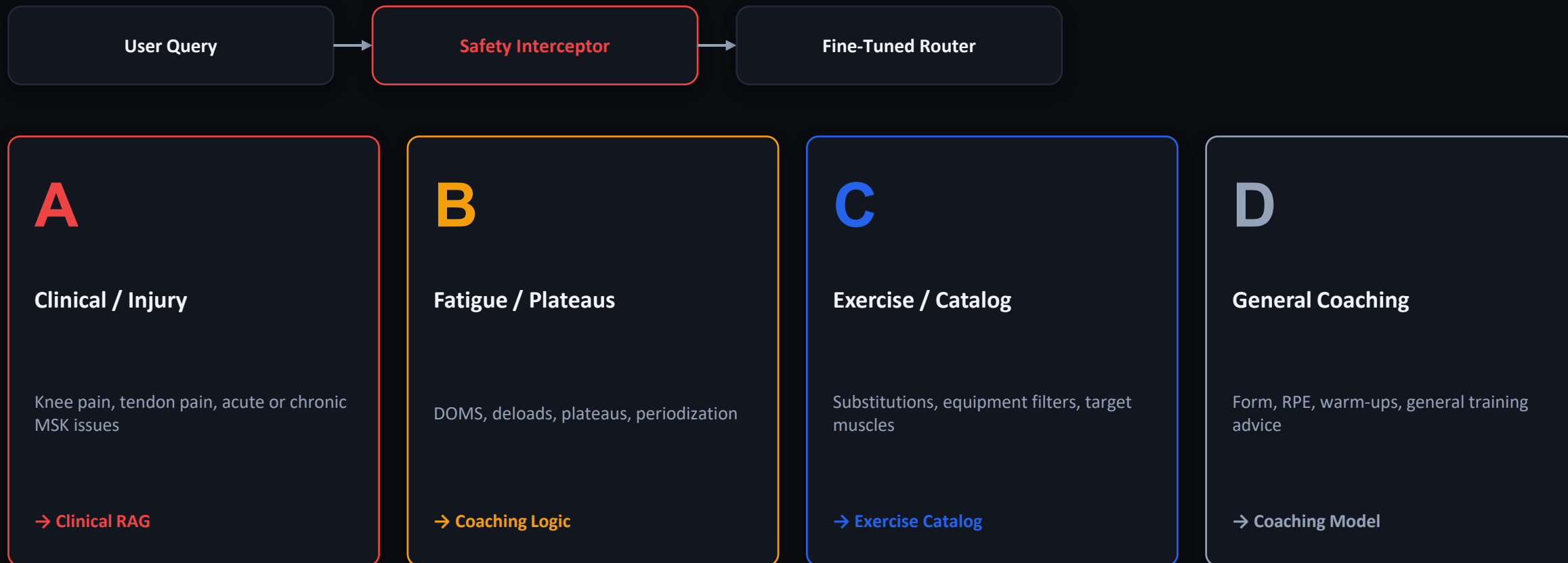


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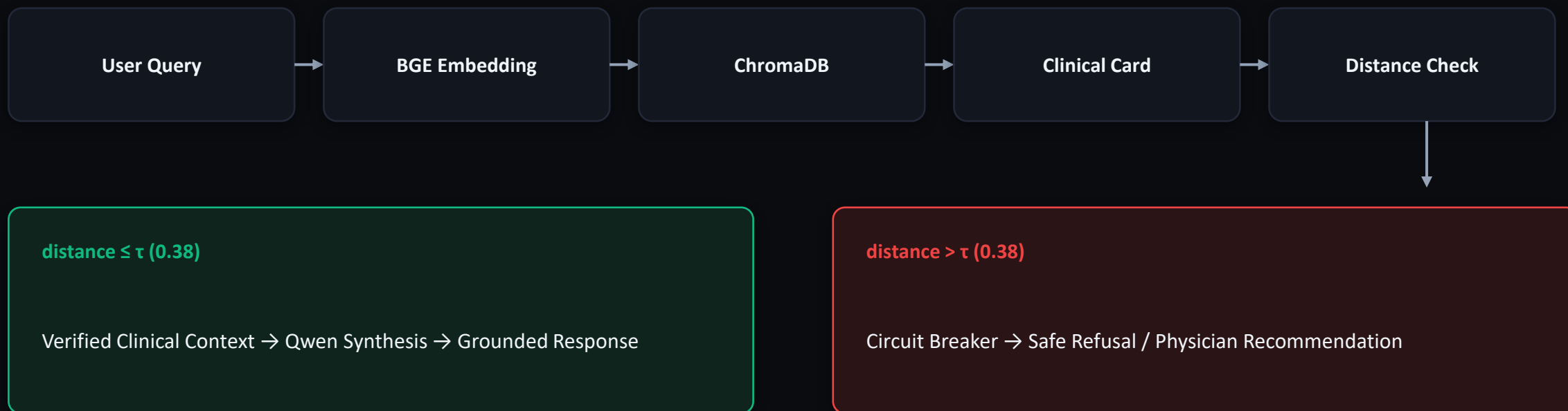
NEGATIVE ANCHORS

Teach the router what NOT to route: non-orthopedic emergencies, PED questions, crash diets, imaging requests, out-of-distribution queries.

The Brain: Intent Routing



Clinical RAG: Grounding the AI



Corpus: 40 clinical cards · Embeddings: BGE-base-en-v1.5 (768-dim) · DB: ChromaDB

Why We Don't Put Everything in One Vector Database

CLINICAL RAG

- 40 canonical clinical cards
- Strict cosine-distance rejection ($\tau = 0.38$)
- Safety-critical semantic retrieval
- Dynamic ingestion, hash-based idempotent updates

VS

PERIODIZATION KNOWLEDGE

- 5 static prompt assets
- Encodes training theory, not clinical facts
- No retrieval / rejection threshold
- Directly injected into Pass 2 synthesis

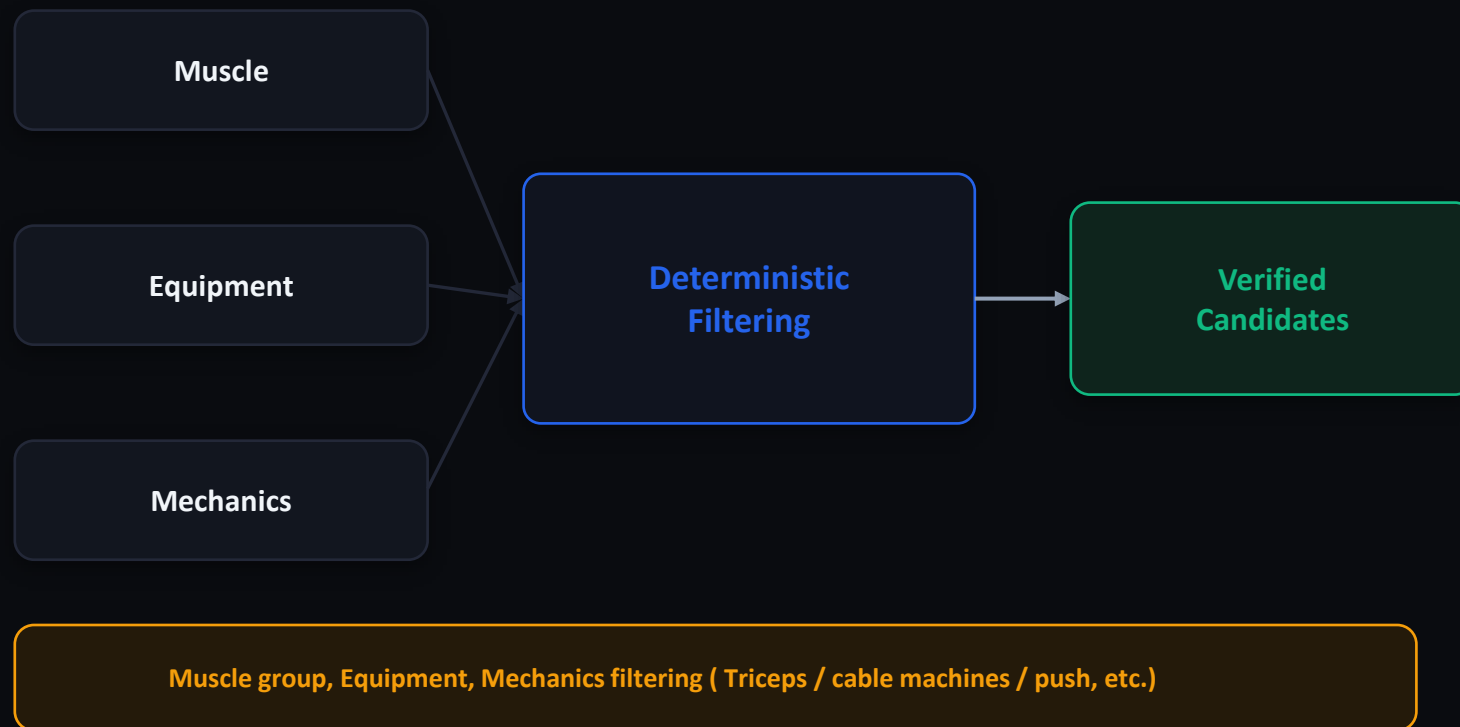
Separation by purpose — not a historical accident, a design decision.

Exercise Intelligence

876

Structured exercises

- ID & Name & Aliases
- Primary / Secondary Muscles
- Equipment
- Mechanics
- Instructions & Images

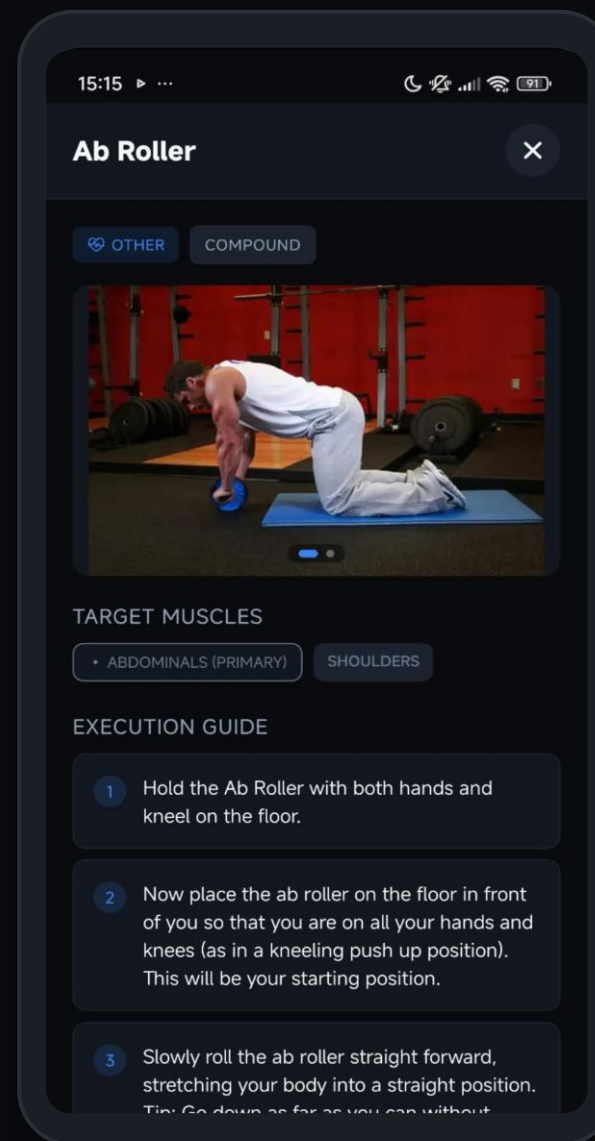


Find the Right Exercise, Instantly

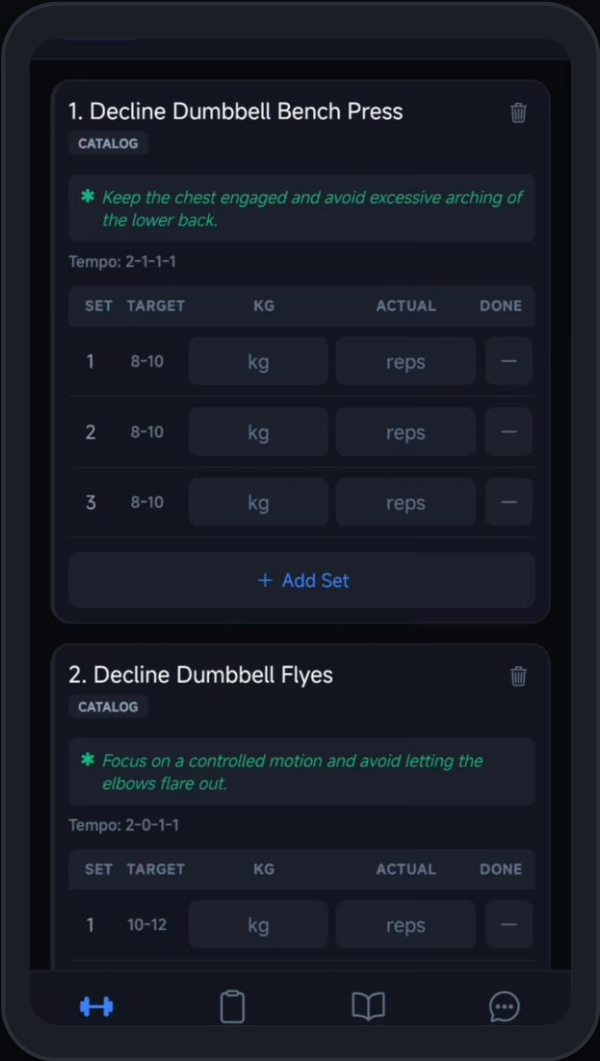
Search by muscle, equipment, or name

Deterministic filters, not guesswork

Browse the full 876-exercise catalog



From Exercise to Workout



Log sets, reps, and weight per exercise



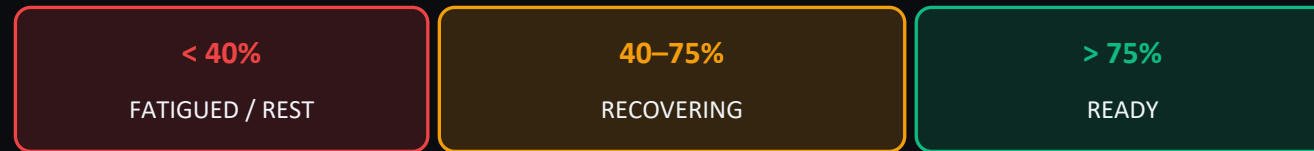
AI-generated workouts as structured JSON



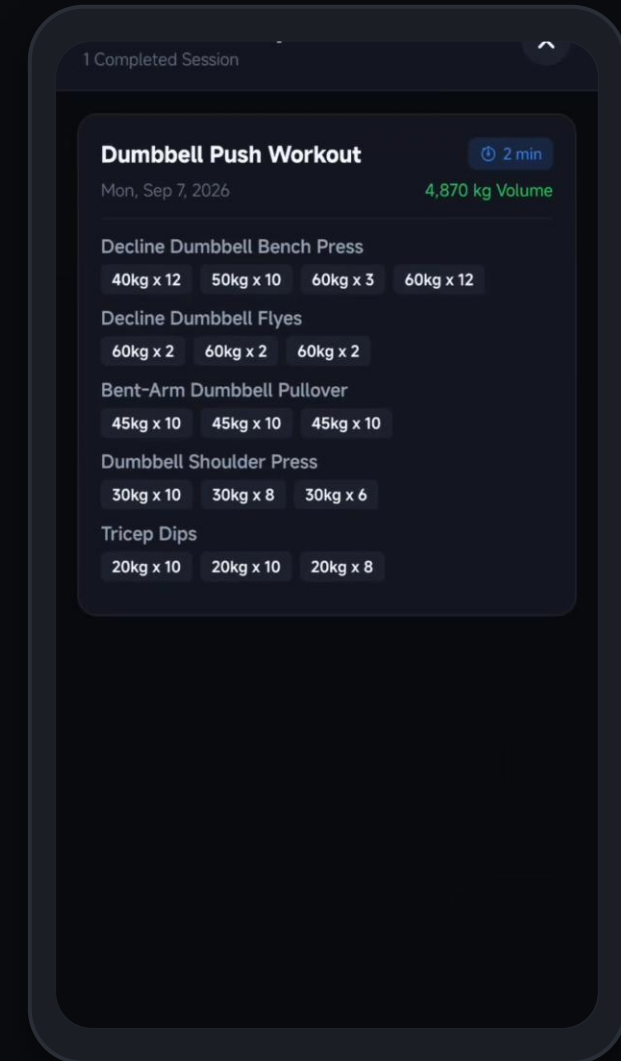
Every exercise ID checked for referential integrity

Recovery Intelligence

Training volume, volume landmarks, and per-muscle recovery drive a single recovery state.

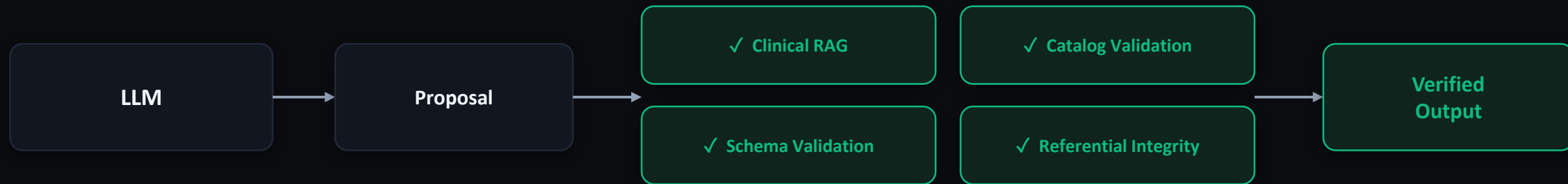


- Per-muscle recovery percentages
- Fatigue vs. readiness state
- Training volume vs. volume landmarks



The Safety Layer

“The LLM proposes. The system verifies.”



The model isn't inherently safe — software controls surround it at every step.

What Broke — And How We Fixed It

1 RAG Consistency PROBLEM Clinical sources can change. FIX SHA-256 idempotent ingestion detects & updates changed docs.	2 Semantic Separation PROBLEM Different knowledge needs different retrieval. FIX Clinical stays in ChromaDB; periodization stays as static assets.	3 Hallucinated Exercise IDs PROBLEM LLM IDs may not exist in the catalog. FIX Referential integrity check against 876 exercises.	4 Safety vs. Intent PROBLEM A query can mix pain signals with a workout ask. FIX Deterministic safety interceptor takes precedence.	5 Limited GPU Resources PROBLEM Large models are costly to serve & tune. FIX QLoRA + 4-bit quantization + Unsloth + vLLM on a T4.
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Let's Put It to the Test

HEALTHY

"Build me a push workout using dumbbells."

Class C → Exercise Catalog → Workout

FATIGUE

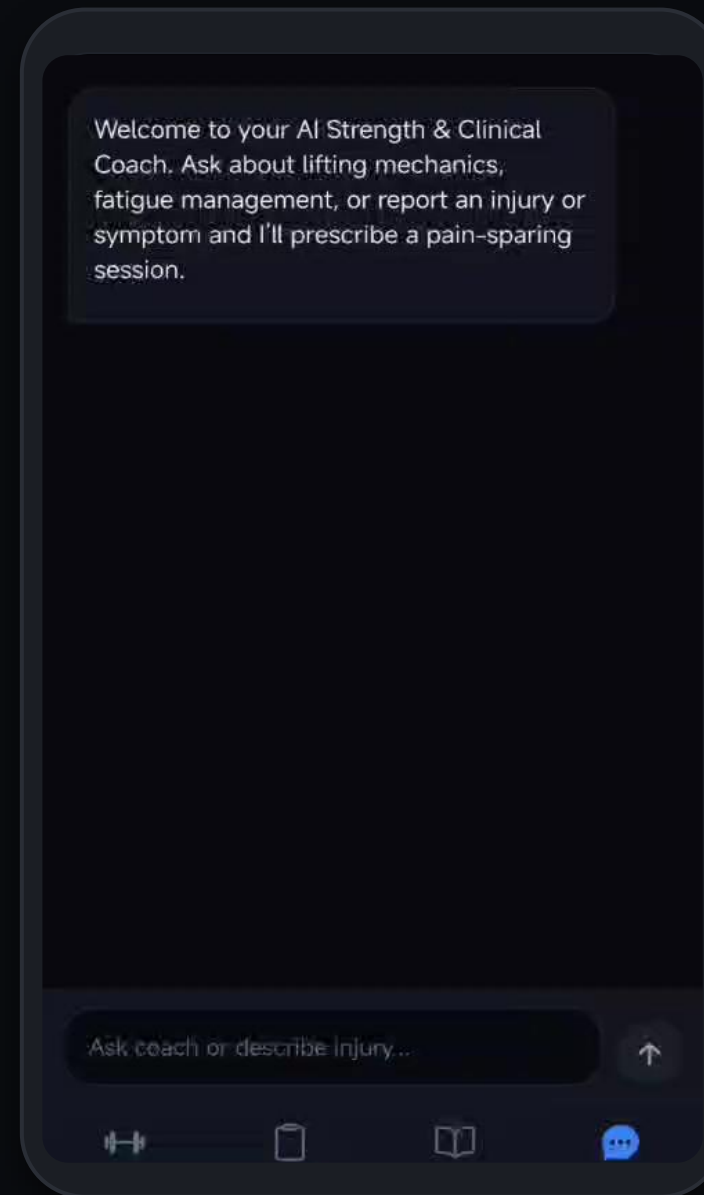
"Severe DOMS after leg day. Should I deload?"

Class B → Coaching Logic

INJURY

"My patellar tendon hurts when I squat. Can I still train legs?"

Safety Interceptor → Class A → Clinical RAG



Where We Go Next

Personalization

Wearables

Computer Vision

Adaptive Programming

Cloud Deployment

Movement-Type Filtering

Clinical → Catalog Integration

Could AI Gym Coach Become a SaaS?

FREE

Basic workouts · Exercise library · Basic tracking

PRO

Personalized AI plans · Recovery analytics · Adaptive plans

PREMIUM / COACH

AI + human coach · Injury-aware programming · Wearables

Customers:

Individual Users

Personal Trainers

Gyms

Sports Academies

Where is the real value — the AI, the data, or the personalization?

OPEN SOURCE

Built to Be Explored



Scan to Explore the Project

GitHub Repository: <https://github.com/AhmedH32/ai-gym-coach.git>